

Q1 (5 points):

From figure 1, we get $P_r(r) = \begin{cases} -2r+2, & r \in [0, 1] \\ 0, & \text{o.w.} \end{cases}$

then $P_r(r) = \int_{-\infty}^r P_r(w) dw = \begin{cases} 0, & r < 0 \\ -r^2+2r, & r \in [0, 1] \\ 1, & r > 1 \end{cases}$

and we have $P_z(z) = \begin{cases} 4z, & z \in [0, \frac{1}{2}] \\ -4z+4, & z \in [\frac{1}{2}, 1] \\ 0, & \text{o.w.} \end{cases}$

the corresponding $P_z(z) = \int_{-\infty}^z P_z(w) dw = \begin{cases} 0, & z < 0 \\ 2z^2, & z \in [0, \frac{1}{2}] \\ -2z^2+4z-1, & z \in [\frac{1}{2}, 1] \\ 1, & z > 1 \end{cases}$

we want to find: $Z(r)$ and let $P_z(z) = P_z(Z(r)) = P_r(r)$

$\Rightarrow Z(r) = P_z^{-1}(P_r(r))$

$\Rightarrow P_z^{-1}(s) = \begin{cases} \sqrt{\frac{s}{2}}, & s \in [0, \frac{1}{2}] \\ 1 - \sqrt{\frac{1-s}{2}}, & s \in [\frac{1}{2}, 1] \\ 0, & \text{o.w.} \end{cases}$

$Z(r) = P_z^{-1}(P_r(r)) = \begin{cases} \sqrt{\frac{-r^2+2r}{2}}, & r \in [0, 1 - \frac{\sqrt{2}}{2}] \\ 1 + \frac{\sqrt{2}}{2}(r-1), & r \in (1 - \frac{\sqrt{2}}{2}, 1] \\ 0, & \text{o.w.} \end{cases}$

Q2 (5 points):

(a) For the 2D convolution:

$$w(x, y) * f(x, y) = \sum_{s=a}^a \sum_{t=b}^b w(s, t) f(x-s, y-t)$$

image size: $m \times n$; kernel size: $a = \frac{m-1}{2}$; $b = \frac{n-1}{2}$

If we have $\sum_{s=a}^a \sum_{t=b}^b w(s, t) = 0$

then we have: $\sum_{x=a+1}^{m+a} \sum_{y=b+1}^{n+b} w(x, y) * f(x, y)$

$$= \sum_{x=a+1}^{m+a} \sum_{y=b+1}^{n+b} \sum_{s=a}^a \sum_{t=b}^b w(s, t) f(x-s, y-t)$$

$$= \sum_{x=1}^m \sum_{y=1}^n f(x, y) \sum_{s=a}^a \sum_{t=b}^b w(s, t)$$

$$= \sum_{x=1}^m \sum_{y=1}^n f(x, y) \cdot 0$$

$$= 0$$

(b) For the 2D correlation:

$$w(x, y) \circ f(x, y) = \sum_{s=a}^a \sum_{t=b}^b w(s, t) f(x+s, y+t)$$

with the same image and kernel size as (a):

If $\sum_{s=a}^a \sum_{t=b}^b w(s, t) = 0$

then: $\sum_{x=a+1}^{m+a} \sum_{y=b+1}^{n+b} w(x, y) \circ f(x, y) = \sum_{x=a+1}^{m+a} \sum_{y=b+1}^{n+b} \sum_{s=a}^a \sum_{t=b}^b w(s, t) f(x+s, y+t)$

$$= \sum_{x=1}^m \sum_{y=1}^n f(x, y) \sum_{s=a}^a \sum_{t=b}^b w(s, t)$$

$$= \sum_{x=1}^m \sum_{y=1}^n f(x, y) \cdot 0$$

$$= 0$$

Q3 (5 points):

$$\text{2D FT: } \begin{cases} F(u, v) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x, y) e^{-j2\pi(ux+vy)} dx dy \\ f(x, y) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} F(u, v) e^{j2\pi(ux+vy)} du dv \end{cases}$$

$$\begin{aligned} \mathcal{F}[f(x, y) * h(x, y)] &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x, y) * h(x, y) e^{-j2\pi(ux+vy)} dx dy \\ &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \left(\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(s, t) h(x-s, y-t) ds dt \right) e^{-j2\pi(ux+vy)} dx dy \\ &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(s, t) \left(\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} h(x-s, y-t) e^{-j2\pi(ux+vy)} dx dy \right) ds dt \\ &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(s, t) \left(\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} h(x, y) e^{-j2\pi(ux+vy)} e^{-j2\pi(us+vt)} dx dy \right) ds dt \\ &= H(u, v) \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(s, t) e^{-j2\pi(us+vt)} ds dt \\ &= F(u, v) H(u, v) \end{aligned}$$

$$\begin{aligned} \mathcal{F}^{-1}[F(u, v) * H(u, v)] &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} F(u, v) * H(u, v) e^{j2\pi(ux+vy)} du dv \\ &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \left(\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} F(s, t) H(u-s, v-t) ds dt \right) e^{j2\pi(ux+vy)} du dv \\ &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} F(s, t) \left(\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} H(u-s, v-t) e^{j2\pi(ux+vy)} du dv \right) ds dt \\ &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} F(s, t) \left(\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} H(u, v) e^{j2\pi(ux+vy)} e^{j2\pi(sx+ty)} du dv \right) ds dt \\ &= h(x, y) \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} F(s, t) e^{j2\pi(sx+ty)} ds dt \\ &= f(x, y) h(x, y) \end{aligned}$$

Then we have

$$f(x, y) * h(x, y) \Leftrightarrow F(u, v) H(u, v)$$

$$f(x, y) h(x, y) \Leftrightarrow F(u, v) * H(u, v)$$

Q4: (5 points)

(a): The spatial mask:

$$h(x, y) = \frac{1}{4} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$h(x, y)$ size is 3×3 ; $f(x, y)$ size is 7×7

The filtering result $g(x, y) = h(x, y) * f(x, y)$

$g(x, y)$ size is 9×9

$$g(x, y) = \begin{bmatrix} 0 & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{2} & 0 & 0 & 0 & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{4} & 0 \\ 0 & 0 & \frac{3}{4} & \frac{3}{4} & \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ 0 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{4} & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

(b): To calculate $H(u, v)$, we need to pad the $h(x, y)$ as the same size of $g(x, y)$, and do the 2D-DFT, $x \in [-4, 4]$, $y \in [-3, 3]$

$$H(u, v) = \sum_{x=-4}^4 \sum_{y=-3}^3 h(x, y) e^{-j2\pi(\frac{ux}{9} + \frac{vy}{7})}$$

$$= \frac{1}{4} [e^{-j2\pi(-\frac{v}{7})} + e^{-j2\pi(\frac{v}{7})} + e^{-j2\pi(-\frac{u}{9})} + e^{-j2\pi(\frac{u}{9})}]$$

$$= \frac{1}{4} [2 \cos(\frac{2}{7}\pi v) + 2 \cos(\frac{2}{9}\pi u)]$$

$$\therefore H(u, v) = \frac{1}{2} [\cos(\frac{2}{9}\pi u) + \cos(\frac{2}{7}\pi v)] \quad \begin{cases} u \in [-4, 4] \text{ and } u \in \mathbb{Z} \\ v \in [-3, 3] \text{ and } v \in \mathbb{Z} \end{cases}$$